

Interpreting Powers of i and the Complex Plane

Answer Key

Algebra 2 • Guided Concept Practice

Quick Answers

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|--------------|----------------------|
| 1. i | 6. $3i, -3i$ |
| 2. $-i$ | 7. $2 - 3i$ |
| 3. $2 + 6i$ | 8. $1 + 2i$ |
| 4. $-4 + 3i$ | 9. $2, -2, 2i, -2i$ |
| 5. 5 | 10. $2 + 3i, 2 - 3i$ |

1. The three empty cells are $i^3 = i^2 \cdot i = -i$, $i^4 = (i^2)^2 = 1$, and $i^5 = i^4 \cdot i = i$.

Since $i^4 = 1$, multiplying by i^4 changes nothing, so the list $i, -1, -i, 1$ repeats forever in blocks of four. That is why i^5 is back at the start:

$$\boxed{i^5 = i}$$

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2. Divide the exponent by the length of the cycle: $27 \div 4 = 6$ remainder 3. Six complete cycles land back at 1, and the remainder says take three more steps: $i^{27} = (i^4)^6 \cdot i^3 = 1 \cdot i^3$, so

$$\boxed{i^{27} = -i}$$

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The remainder is the whole method: remainder 0 $\rightarrow 1$, 1 $\rightarrow i$, 2 $\rightarrow -1$, 3 $\rightarrow -i$.

3. The arrows are $(3, 2)$ and $(-1, 4)$. Sliding the second arrow to the tip of the first moves it right -1 and up 4 from the point $(3, 2)$, landing at $(2, 6)$. In notation that is exactly adding the parts separately: $(3 + 2i) + (-1 + 4i) = (3 - 1) + (2 + 4)i$, so

$$\boxed{2 + 6i}$$

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The picture and the algebra are the same statement: complex addition is vector addition, componentwise.

4. $i(3 + 4i) = 3i + 4i^2 = 3i + 4(-1) = -4 + 3i$, so

$$\boxed{-4 + 3i}$$

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On the grid the arrow to $(3, 4)$ becomes the arrow to $(-4, 3)$: same length, turned a quarter turn (90°) counterclockwise about the origin. Accept any sentence saying the arrow rotated 90° counterclockwise without changing length. That is what multiplying by i means geometrically, and it explains the cycle in problem 1: four quarter turns return you to where you started, which is $i^4 = 1$.

5. The point $(3, 4)$ sits 3 across and 4 up, so the right triangle dropped to the real axis has legs 3 and 4. By the Pythagorean theorem the distance from the origin is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25}$, giving

$$\boxed{5}$$

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This distance is called the modulus, written $|3 + 4i|$.

6. $x^2 + 9 = 0$ gives $x^2 = -9$, so $x = \pm\sqrt{-9} = \pm 3i$.

$$\boxed{x = 3i \quad \text{or} \quad x = -3i}$$

Both solutions sit on the imaginary axis, at $(0, 3)$ and $(0, -3)$: same distance from the origin, opposite directions. The parabola $y = x^2 + 9$ never crosses the x -axis precisely because neither solution has a real part to land on — the solutions exist, but not on the real number line the graph is drawn over.

7. Reduce each power with the remainder rule. $15 \div 4$ leaves remainder 3, so $i^{15} = -i$. $22 \div 4$ leaves remainder 2, so $i^{22} = -1$. Substituting: $3i^{15} - 2i^{22} = 3(-i) - 2(-1) = -3i + 2$, so

$$\boxed{2 - 3i}$$

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Watch the second sign: subtracting -1 adds, which is where this problem is usually lost.

8. $i^4 = 1$, so $i^4(1 + 2i) = 1(1 + 2i)$, giving

$$\boxed{1 + 2i}$$

— you land exactly where you started.

The picture says it first: each multiplication by i turns the arrow a quarter turn, so four of them is a full 360° turn and every point returns to itself. The arithmetic only confirms what the rotation already guaranteed.

9. Factor as a difference of squares twice: $x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$. The first two factors give real zeros; $x^2 + 4 = 0$ gives $x^2 = -4$, so $x = \pm 2i$.

$$\boxed{x = 2, -2, 2i, -2i}$$

The real zeros are 2 and -2 ; the non-real ones are $2i$ and $-2i$. Plotted, the four points sit at the four compass points of a circle of radius 2 about the origin — one every quarter turn, which is the same fourfold symmetry the powers of i have.

10. Completing the square: $x^2 - 4x + 13 = (x - 2)^2 + 9 = 0$, so $(x - 2)^2 = -9$ and $x - 2 = \pm 3i$.

$$\boxed{x = 2 + 3i \quad \text{or} \quad x = 2 - 3i}$$

(The quadratic formula gives the same thing: the discriminant is $16 - 52 = -36$, and $\sqrt{-36} = 6i$.)

The two points are $(2, 3)$ and $(2, -3)$ — reflections of each other across the real axis. This must happen whenever the coefficients are real: the quadratic formula adds and subtracts the same $\sqrt{\text{negative}}$ term, so the two answers share a real part and carry opposite imaginary parts. Non-real solutions of a real polynomial always come in such conjugate pairs, which is also why problem 9 produced $2i$ and $-2i$ together.